

TrES-5b

R-0871

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_240918 · created 2026-07-28T23:30:44+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460571.8194 +/- 0.0039 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.05
Transit depth (Rp/Rs)^2	3.37 +/- 1.83 [%]
Orbital Inclination (inc)	89.16 +/- 2.59
Airmass coefficient 1 (a1)	0.896 +/- 0.06
Airmass coefficient 2 (a2)	0.053 +/- 0.041
Scatter in the residuals of the lightcurve fit is	6.83 %
Best Comparison Star	#3 - [548, 86]
Optimal Aperture	3.23
Optimal Annulus	5.91
Transit Duration (day)	0.0881 +/- 0.0056

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.184 ± 0.05 vs 0.142 (deviation 0.84σ)
Duration fit vs published	0.0881 d vs 0.0758 d (deviation 2.19σ)
Mid-time O–C	-10.2 min
Depth SNR	3.7
Residual scatter	6.83 %

Light curve

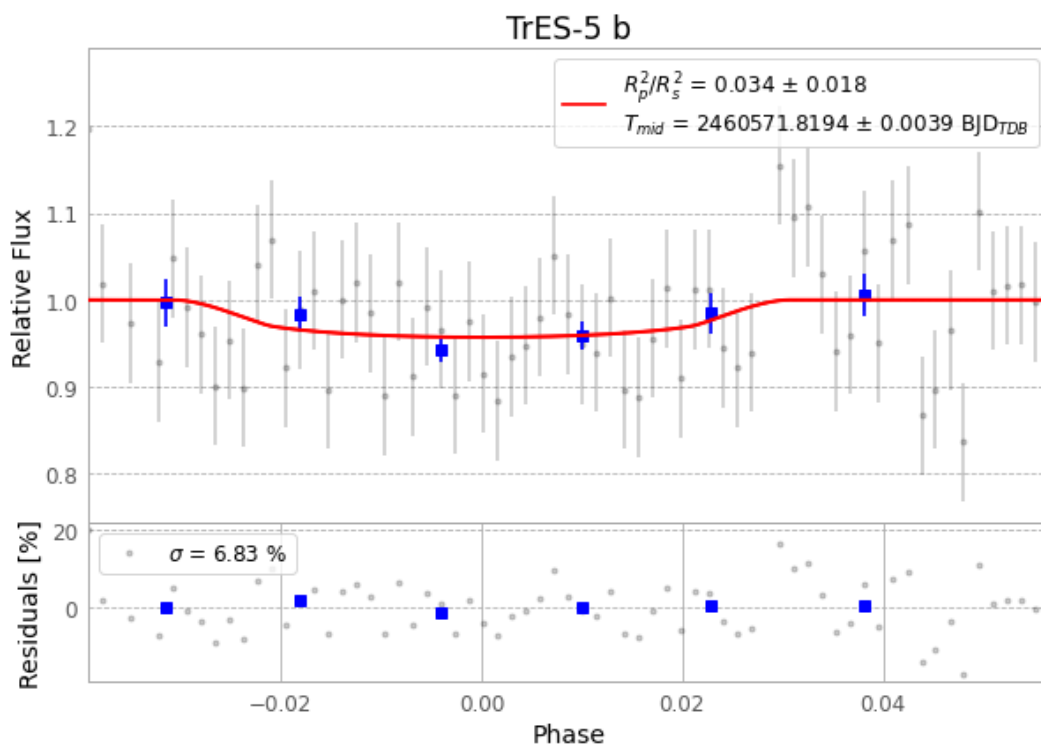


Figure 1 — FinalLightCurve_TrES-5 b_2024-09-17.png (SHA-256 01293835bc75c8d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 290], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 7bc53f0e70de5397...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

71 FITS frames (47 MB), TRES-5240918060615.FITS → TRES-5240918093615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-240918.log (1bd717bc1596...), FinalLightCurve_TrES-5 b_2024-09-17.png (01293835bc75...), FinalLightCurve_TrES-5 b_2024-09-17.csv (2bf9e506daa8...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 23:30Z.